10 · Case 6, part 2: glucagon, the day graph, and the tired pancreas

The words of this track, as it says them. Bold brown is the examiner.

Exam case 6, continued. Glucagon, the blood sugar day, and the tired pancreas.

These five aren't starred, but they sit inside the yellow case, and her deck answers every one of them, on slides thirty-one to thirty-four. Her helpline says an unstarred question can still be asked.

Three marks. What other molecules also influence blood glucose levels?

Glucagon, and it raises blood glucose three ways. It stimulates the breakdown of glycogen: glycogenolysis. It stimulates the breakdown of fats: lipolysis. And it stimulates the liver to make new glucose from non-glucose sources: gluconeogenesis.

Four marks. Insulin is designed to only be high after a meal, so in normal function it follows a pattern like the diagram. Can you describe the normal blood glucose level over the day? What about insulin?

Blood glucose changes over the day with your meals, especially carbs and sugar: it rises after each meal. But there's always a background level of insulin. Insulin is never at zero. Blood glucose is never at zero either: it stays within the normal range. And her two words for insulin: basal insulin is the background level, and bolus insulin is what's released, or given, after meals.

One mark. In normal blood sugar regulation, does the blood glucose level go above the normal range?

No. It shouldn't exceed the normal range.

Two marks. What happens overnight? And what could that mean for the morning?

Blood glucose drops to its lowest overnight. And breakfast is breaking the fast: you wake up hungry, because your body has run out of glycogen.

Three marks. When the body's cells are resistant to the insulin signal, the pancreas tries to compensate by making more insulin. All that overwork can damage it, so now it struggles to make enough. What would happen in the body now, and how might we need to intervene?

Pancreatic fatigue: damage to the pancreas from overworking. Then low insulin levels, so blood glucose regulation is impaired. That's type 2 diabetes mellitus, and the person might end up on insulin. Which is exactly where Larry is.

Once more. Glucagon, and the day.

Glucagon raises blood glucose three ways: glycogenolysis, lipolysis, and gluconeogenesis in the liver. Over the day, blood glucose follows your meals, but it never hits zero and stays in the normal range, and insulin never hits zero either. Basal insulin is the background; bolus insulin comes after meals. Normally, blood glucose never goes above the normal range. It's lowest overnight, so breakfast breaks the fast. Insulin resistance overworks the pancreas until it fatigues and makes too little insulin: type 2 diabetes, and possibly insulin injections.